

Mohammed Thaha Dawood

Backend Software Engineer | Node.js | Java | Distributed Systems | AWS

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SUMMARY

Backend Software Engineer with 2.5 years building scalable distributed systems and high-throughput REST APIs on AWS. Reduced p95 latency **40%**, DB query time **75%**, and cloud costs **25%** on a production multi-tenant SaaS ERP serving 20+ B2B tenants. Author of 4 open-source backend infrastructure libraries. Proficient in Node.js/NestJS and Java/Spring Boot.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Java, Python, SQL
Backend	Node.js, NestJS, Express.js, Spring Boot, Spring MVC, Spring Data JPA, Hibernate, Flask
Databases	MongoDB, PostgreSQL, MySQL, Redis (caching, Lua scripting, pub/sub), Mongoose
Messaging	BullMQ, Apache Kafka, RabbitMQ, event-driven architecture, transactional outbox
Cloud	AWS (ECS, EC2, S3, CloudWatch, IAM, VPC), Docker, Docker Compose, Kubernetes, Jenkins, Nginx
Patterns	Microservices, SAGA orchestration, rate limiting, idempotency, CQRS, multi-tenancy, distributed locking
Auth	JWT, OAuth2, HMAC, RBAC, Passport.js, API security, tenant isolation
Testing	Jest, JUnit 5, Mockito, Testcontainers, integration testing, concurrency testing, TDD
Tools	Git, Swagger/OpenAPI, Postman, CI/CD, Jira, IntelliJ IDEA, VS Code

EXPERIENCE

Software Engineer — *CYGNOSZ (Cygnonex), Kochi* **Mar 2024 – Present**

- Owned backend architecture of a **production multi-tenant SaaS ERP** (4+ microservices, AWS ECS) serving 20+ B2B tenants; led 5 engineers across system design, development, and deployment — delivering all milestones on schedule.
- Reduced **p95 API latency 40%** (~800ms → ~480ms) via N+1 elimination, compound indexing, and Redis caching; independently cut critical query time **75%** through targeted index strategy.
- Engineered **idempotent billing workflows** with DB transactions and distributed locks — exactly-once execution across 1,000+ daily financial operations under concurrent load and partial failures.
- Enforced **multi-tenant data isolation** across data access layers, JWT/OAuth2 auth, and RBAC — zero cross-tenant leakage incidents across shared infrastructure.
- Built **async job pipelines** (BullMQ/Redis) handling 5,000+ daily jobs for reports, notifications, and sync; tuned worker concurrency to prevent queue saturation under peak load.
- Drove **25% AWS cost reduction** via ECS task rightsizing and architecture refactoring; established CloudWatch dashboards and alerting for production observability.
- Enabled reliable billing and zero-downtime operations for 20+ SaaS tenants by combining exactly-once payment workflows, failure-resilient async pipelines, and versioned API rollouts.

Backend Developer Intern — *Lexyle Technologies, Kerala* **Sep 2023 – Feb 2024**

- Built and shipped REST APIs in Node.js/Express.js for internal SaaS tooling used by 3+ clients; contributed to MySQL schema design and query optimisation under senior engineer guidance.
- Integrated Docker-based CI/CD pipelines and wrote unit tests, reducing manual deployment steps and improving release reliability across the team.

OPEN-SOURCE PROJECTS

sentinel-gateway (*Java 21, Spring Boot 3, Redis, PostgreSQL, Docker*) github.com/MohdThaha/Sentinel-gateway

- API gateway with pluggable rate limiting — Token Bucket and Sliding Window Counter via atomic Redis Lua scripts; strategy switchable per-client via REST admin API with zero code change.

outrelay (*NestJS, MongoDB, BullMQ, Redis, TypeScript*) github.com/MohdThaha/outrelay

- Solves silent event loss — Transactional Outbox Pattern with MongoDB change streams, HMAC-signed webhooks, and at-least-once delivery surviving crashes and network partitions.

saga-orchestrator (*NestJS, BullMQ, Redis, TypeScript*) github.com/MohdThaha/saga-orchestrator

- Prevents ghost state in distributed transactions — SAGA orchestrator with automatic compensating transactions, configurable retries, and idempotency headers for safe step re-execution.

idem-guard (*NestJS, Redis, MongoDB, TypeScript*) github.com/MohdThaha/Idem-guard

- Eliminates duplicate execution from retries and race conditions — `@Idempotent()` decorator; atomic Redis `SET NX EX` locking, TTL response cache, MongoDB audit log, REST invalidation API.

EDUCATION

B.Tech, Computer Science & Engineering — *Valia Koonambaikulathamma College of Engineering & Technology* 2019 – 2023